

# Advanced Excel & MIS

## Detailed Chapter Notes & Worked Examples - Volume 2

This pack develops practical understanding through explanations, worked examples, lab tasks and review questions. Complete the matching Foundation Pack first.

| Chapter | Topic                |
|---------|----------------------|
| 1       | Reliable formulas    |
| 2       | Lookup and cleaning  |
| 3       | Pivot MIS            |
| 4       | Audit and protection |

# Chapter 1: Reliable formulas

## Concept notes

Use absolute references for fixed values, structured references for tables and IFERROR only after understanding the likely error. Separate input, calculation and report areas.

## Worked example

If Amount is in B2 and GST rate in \$F\$1, tax is =B2\*\$F\$1. A conditional total is =SUMIFS(Sales[Amount],Sales[Month],H2,Sales[Branch],H3).

## Lab assignment

Create a 50-row sales table and calculate tax, net value and branch-month totals.

## Review questions

1. Difference between relative and absolute reference?
2. When should SUMIFS be used?
3. Why should inputs be separated?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

## Chapter 2: Lookup and cleaning

### Concept notes

XLOOKUP retrieves related data by a key. TRIM removes extra spaces; CLEAN removes non-printing characters; VALUE converts numeric text. Validate uniqueness before lookup.

### Worked example

=XLOOKUP(A2,Students[ID],Students[Name],"Not found") returns the student name and handles a missing ID.

### Lab assignment

Clean a deliberately untidy 30-row list and connect it to a master table.

### Review questions

1. What makes a good lookup key?
2. Why can numeric text break reports?
3. What does the not-found argument do?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

## Chapter 3: Pivot MIS

### Concept notes

A PivotTable groups large records without writing many formulas. Correct source structure requires one header row, no merged cells and consistent data types.

### Worked example

Place Branch in Rows, Month in Columns and Amount in Values; add a slicer for Product.

### Lab assignment

Build a fees dashboard with PivotTable, chart, slicers and four KPI cards.

### Review questions

1. Why avoid blank header names?
2. Difference between filter and slicer?
3. When should source data be refreshed?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

## Chapter 4: Audit and protection

### Concept notes

Trace precedents, evaluate formulas and reconciliation totals help detect errors. Protect formulas but keep input cells editable. Never hide an unexplained balancing difference.

### Worked example

Opening + Receipts - Payments should equal Closing. Add a check cell that displays OK only when difference is zero.

### Lab assignment

Add validation, protection, error checks and a control sheet to your dashboard.

### Review questions

1. What is a control total?
2. Why is hiding errors dangerous?
3. What should remain editable?

Evidence to submit: source file or working output, screenshot/PDF, and a 100-word explanation of decisions and checks.

## Final Applied Assignment

Create one original project that combines all four chapters of Advanced Excel & MIS. Use realistic but non-sensitive sample data. Submit planning notes, source files, final output, test evidence and a one-page reflection.

### Assessment rubric

| Area                               | Marks |
|------------------------------------|-------|
| Concept accuracy                   | 25    |
| Working practical output           | 30    |
| Validation and testing             | 20    |
| File organisation and presentation | 15    |
| Original explanation               | 10    |

### Student declaration

I confirm that the submitted work is my own practice. I have not included passwords, OTPs, identity documents or confidential student records.

Support: [mcieducationalgroup@gmail.com](mailto:mcieducationalgroup@gmail.com) | 9334779133, 7004773247